

sales-analysis

0.0.1 Import Necessary Libraries

```
[1]: import pandas as pd
import os
import matplotlib.pyplot as plt
```

0.0.2 Merge 12 Month of Sales Data into a Single CSV File

```
[2]: files = [file for file in os.listdir("./Sales_Data")]
for f in files:
    print(f)
```

Sales_April_2019.csv
Sales_August_2019.csv
Sales_December_2019.csv
Sales_February_2019.csv
Sales_January_2019.csv
Sales_July_2019.csv
Sales_June_2019.csv
Sales_March_2019.csv
Sales_May_2019.csv
Sales_November_2019.csv
Sales_October_2019.csv
Sales_September_2019.csv

```
[3]: df_list = []

for dir in files:
    file_path = os.path.join("./Sales_Data", dir)    # use to combine multiple
    ↪ parts of a file path
    df = pd.read_csv(file_path)
    df_list.append(df)

# concate all dataframes
combined_df = pd.concat(df_list, ignore_index=True)

# save all files in one file
combined_df.to_csv("all_data.csv", index=False)
```

```
[4]: all_data = pd.read_csv("all_data.csv")
all_data.head()
```

```
[4]:
```

	Order ID	Product	Quantity Ordered	Price Each	\
0	176558	USB-C Charging Cable	2	11.95	
1	NaN	NaN	NaN	NaN	
2	176559	Bose SoundSport Headphones	1	99.99	
3	176560	Google Phone	1	600	
4	176560	Wired Headphones	1	11.99	

	Order Date	Purchase Address
0	04/19/19 08:46	917 1st St, Dallas, TX 75001
1	NaN	NaN
2	04/07/19 22:30	682 Chestnut St, Boston, MA 02215
3	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001
4	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001

0.0.3 Clean up the Data

```
[5]: all_data.isnull().sum()
```

```
[5]:
```

Order ID	545
Product	545
Quantity Ordered	545
Price Each	545
Order Date	545
Purchase Address	545

dtype: int64

```
[6]: all_data = all_data.dropna(how="all")
```

```
[7]: all_data.isnull().sum()
```

```
[7]:
```

Order ID	0
Product	0
Quantity Ordered	0
Price Each	0
Order Date	0
Purchase Address	0

dtype: int64

Find “OR” and dlt it

```
[8]: all_data.columns
```

```
[8]: Index(['Order ID', 'Product', 'Quantity Ordered', 'Price Each', 'Order Date',
          'Purchase Address'],
```

```
dtype='object')
```

```
[9]: all_data = all_data[all_data["Order Date"].str[0:2] != "Or"]
```

Convert Column to the Correct Type

```
[10]: all_data["Quantity Ordered"] = pd.to_numeric(all_data["Quantity Ordered"]) #_
      ↪make int
      all_data["Price Each"] = pd.to_numeric(all_data["Price Each"]) #_
      ↪make float
```

Add Month Column

```
[11]: all_data["Month"] = all_data["Order Date"].str[0:2]
      all_data["Month"] = all_data["Month"].astype("int32")
      all_data.head()
```

```
[11]:
```

	Order ID	Product	Quantity Ordered	Price Each	\
0	176558	USB-C Charging Cable	2	11.95	
2	176559	Bose SoundSport Headphones	1	99.99	
3	176560	Google Phone	1	600.00	
4	176560	Wired Headphones	1	11.99	
5	176561	Wired Headphones	1	11.99	

	Order Date	Purchase Address	Month
0	04/19/19 08:46	917 1st St, Dallas, TX 75001	4
2	04/07/19 22:30	682 Chestnut St, Boston, MA 02215	4
3	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001	4
4	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001	4
5	04/30/19 09:27	333 8th St, Los Angeles, CA 90001	4

Add Sales Column

```
[12]: all_data["Sales"] = all_data["Quantity Ordered"] * all_data["Price Each"]
      all_data
```

```
[12]:
```

	Order ID	Product	Quantity Ordered	Price Each	\
0	176558	USB-C Charging Cable	2	11.95	
2	176559	Bose SoundSport Headphones	1	99.99	
3	176560	Google Phone	1	600.00	
4	176560	Wired Headphones	1	11.99	
5	176561	Wired Headphones	1	11.99	
...	...	...	...	...	
186845	259353	AAA Batteries (4-pack)	3	2.99	
186846	259354	iPhone	1	700.00	
186847	259355	iPhone	1	700.00	
186848	259356	34in Ultrawide Monitor	1	379.99	
186849	259357	USB-C Charging Cable	1	11.95	

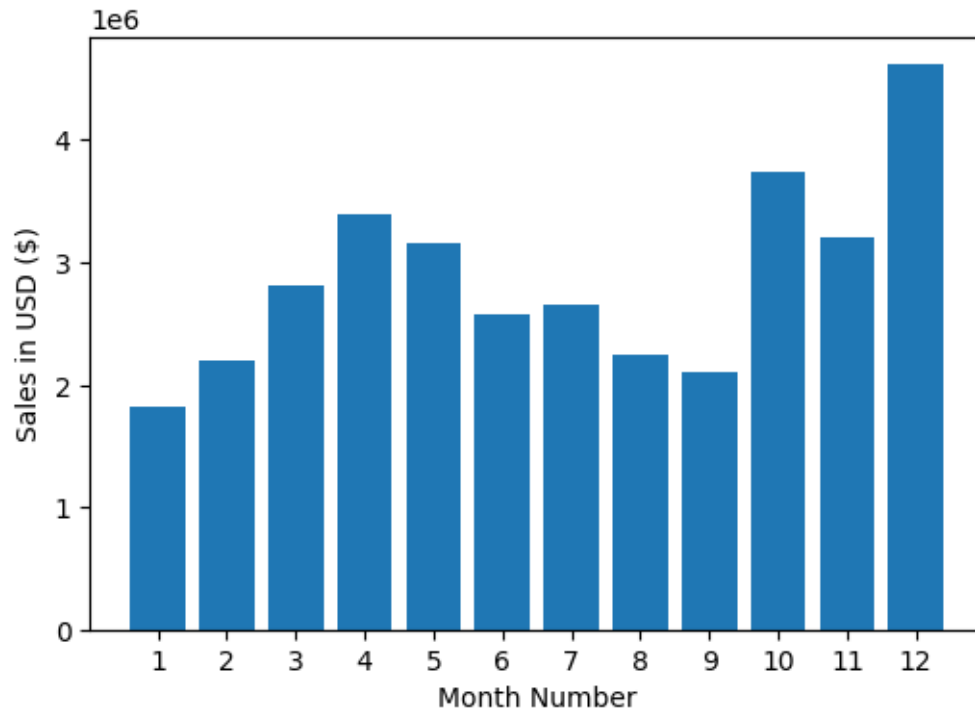
	Order Date	Purchase Address	Month	Sales
0	04/19/19 08:46	917 1st St, Dallas, TX 75001	4	23.90
2	04/07/19 22:30	682 Chestnut St, Boston, MA 02215	4	99.99
3	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001	4	600.00
4	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001	4	11.99
5	04/30/19 09:27	333 8th St, Los Angeles, CA 90001	4	11.99
...	...	...	...	...
186845	09/17/19 20:56	840 Highland St, Los Angeles, CA 90001	9	8.97
186846	09/01/19 16:00	216 Dogwood St, San Francisco, CA 94016	9	700.00
186847	09/23/19 07:39	220 12th St, San Francisco, CA 94016	9	700.00
186848	09/19/19 17:30	511 Forest St, San Francisco, CA 94016	9	379.99
186849	09/30/19 00:18	250 Meadow St, San Francisco, CA 94016	9	11.95

[185950 rows x 8 columns]

0.0.4 Which is the Best Month for Sales?

```
[13]: results = all_data.groupby("Month").sum()
```

```
[14]: plt.figure(figsize=(6,4))
months = range(1,13)
plt.bar(months, results["Sales"])
plt.xticks(months)
plt.xlabel("Month Number")
plt.ylabel("Sales in USD ($)")
plt.show()
```



Add City Column

```
[15]: def get_city(address):
        return address.split(",")[1]

def get_state(address):
    return address.split(",")[2].split(" ")[1]

all_data["City"] = all_data["Purchase Address"].apply(lambda x: f"{get_city(x)}\n↪({get_state(x)})")
all_data
```

```
[15]:
```

	Order ID	Product	Quantity Ordered	Price Each \
0	176558	USB-C Charging Cable	2	11.95
2	176559	Bose SoundSport Headphones	1	99.99
3	176560	Google Phone	1	600.00
4	176560	Wired Headphones	1	11.99
5	176561	Wired Headphones	1	11.99
...	...	...	...	...
186845	259353	AAA Batteries (4-pack)	3	2.99
186846	259354	iPhone	1	700.00
186847	259355	iPhone	1	700.00
186848	259356	34in Ultrawide Monitor	1	379.99
186849	259357	USB-C Charging Cable	1	11.95

	Order Date	Purchase Address	Month	\
0	04/19/19 08:46	917 1st St, Dallas, TX 75001	4	
2	04/07/19 22:30	682 Chestnut St, Boston, MA 02215	4	
3	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001	4	
4	04/12/19 14:38	669 Spruce St, Los Angeles, CA 90001	4	
5	04/30/19 09:27	333 8th St, Los Angeles, CA 90001	4	
...	...	...	...	
186845	09/17/19 20:56	840 Highland St, Los Angeles, CA 90001	9	
186846	09/01/19 16:00	216 Dogwood St, San Francisco, CA 94016	9	
186847	09/23/19 07:39	220 12th St, San Francisco, CA 94016	9	
186848	09/19/19 17:30	511 Forest St, San Francisco, CA 94016	9	
186849	09/30/19 00:18	250 Meadow St, San Francisco, CA 94016	9	

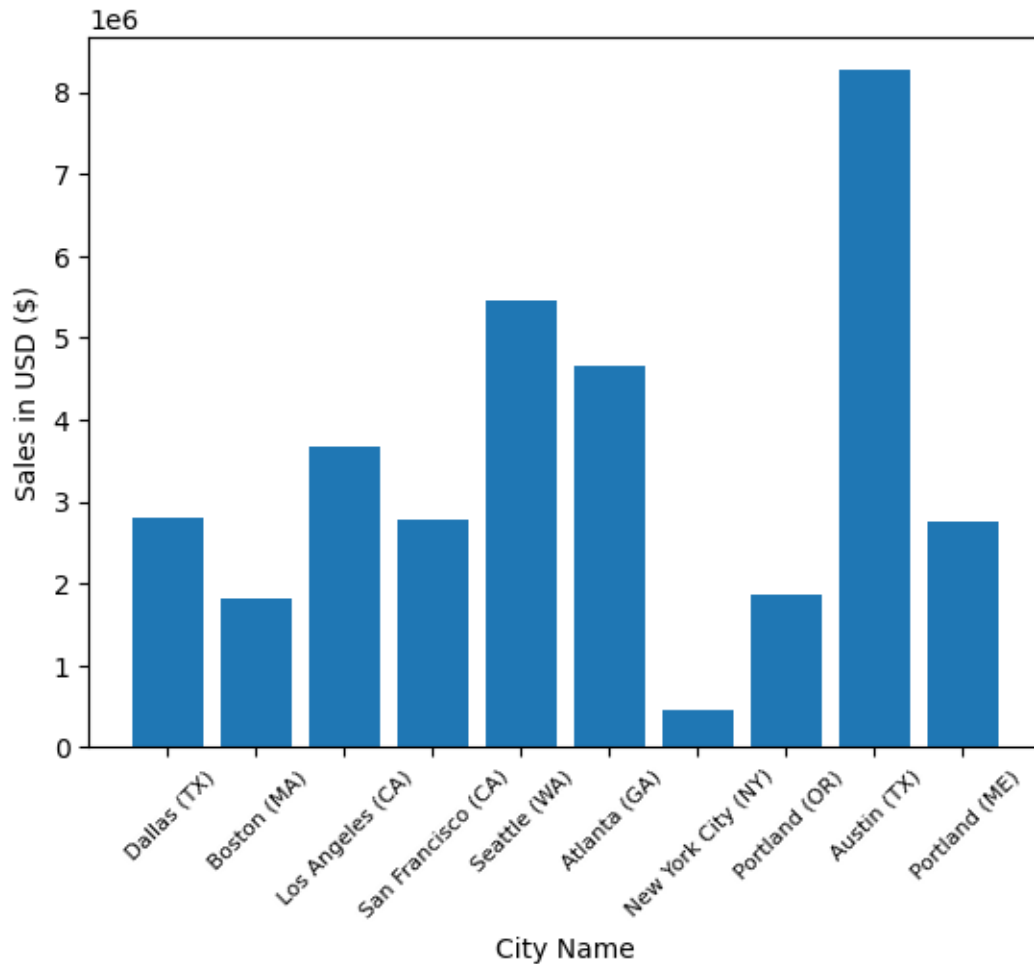
	Sales	City
0	23.90	Dallas (TX)
2	99.99	Boston (MA)
3	600.00	Los Angeles (CA)
4	11.99	Los Angeles (CA)
5	11.99	Los Angeles (CA)
...	...	...
186845	8.97	Los Angeles (CA)
186846	700.00	San Francisco (CA)
186847	700.00	San Francisco (CA)
186848	379.99	San Francisco (CA)
186849	11.95	San Francisco (CA)

[185950 rows x 9 columns]

0.0.5 Which City had the Highest Sales?

```
[16]: results = all_data.groupby("City").sum()
```

```
[17]: cities = all_data["City"].unique()
plt.bar(cities, results["Sales"])
plt.xticks(cities, rotation=45, size=8)
plt.xlabel("City Name")
plt.ylabel("Sales in USD ($)")
plt.show()
```



0.0.6 What time should we display advertisements to maximize the likelihood of customers buying the product?

```
[18]: all_data["Order Date"] = pd.to_datetime(all_data["Order Date"]) # convert_
      ↪datetime format
```

C:\Users\DELL\AppData\Local\Temp\ipykernel_9116\2228339044.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
all_data["Order Date"] = pd.to_datetime(all_data["Order Date"]) # convert
datetime format
```

```
[19]: all_data.head()
```

```
[19]:
```

	Order ID	Product	Quantity Ordered	Price Each	\
0	176558	USB-C Charging Cable	2	11.95	
2	176559	Bose SoundSport Headphones	1	99.99	
3	176560	Google Phone	1	600.00	
4	176560	Wired Headphones	1	11.99	
5	176561	Wired Headphones	1	11.99	

	Order Date	Purchase Address	Month	Sales	\
0	2019-04-19 08:46:00	917 1st St, Dallas, TX 75001	4	23.90	
2	2019-04-07 22:30:00	682 Chestnut St, Boston, MA 02215	4	99.99	
3	2019-04-12 14:38:00	669 Spruce St, Los Angeles, CA 90001	4	600.00	
4	2019-04-12 14:38:00	669 Spruce St, Los Angeles, CA 90001	4	11.99	
5	2019-04-30 09:27:00	333 8th St, Los Angeles, CA 90001	4	11.99	

	City
0	Dallas (TX)
2	Boston (MA)
3	Los Angeles (CA)
4	Los Angeles (CA)
5	Los Angeles (CA)

```
[20]: all_data["Hour"] = all_data["Order Date"].dt.hour
all_data["Minute"] = all_data["Order Date"].dt.minute
```

```
[21]: all_data.head()
```

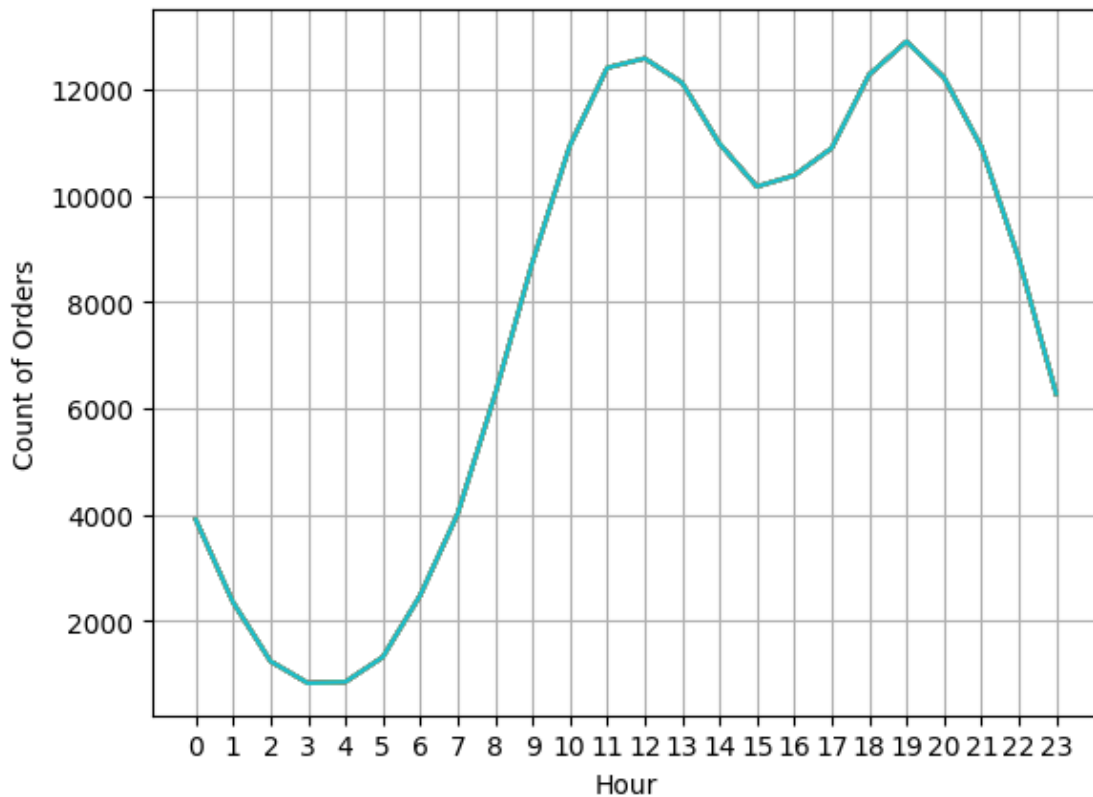
```
[21]:
```

	Order ID	Product	Quantity Ordered	Price Each	\
0	176558	USB-C Charging Cable	2	11.95	
2	176559	Bose SoundSport Headphones	1	99.99	
3	176560	Google Phone	1	600.00	
4	176560	Wired Headphones	1	11.99	
5	176561	Wired Headphones	1	11.99	

	Order Date	Purchase Address	Month	Sales	\
0	2019-04-19 08:46:00	917 1st St, Dallas, TX 75001	4	23.90	
2	2019-04-07 22:30:00	682 Chestnut St, Boston, MA 02215	4	99.99	
3	2019-04-12 14:38:00	669 Spruce St, Los Angeles, CA 90001	4	600.00	
4	2019-04-12 14:38:00	669 Spruce St, Los Angeles, CA 90001	4	11.99	
5	2019-04-30 09:27:00	333 8th St, Los Angeles, CA 90001	4	11.99	

	City	Hour	Minute
0	Dallas (TX)	8	46
2	Boston (MA)	22	30
3	Los Angeles (CA)	14	38
4	Los Angeles (CA)	14	38
5	Los Angeles (CA)	9	27

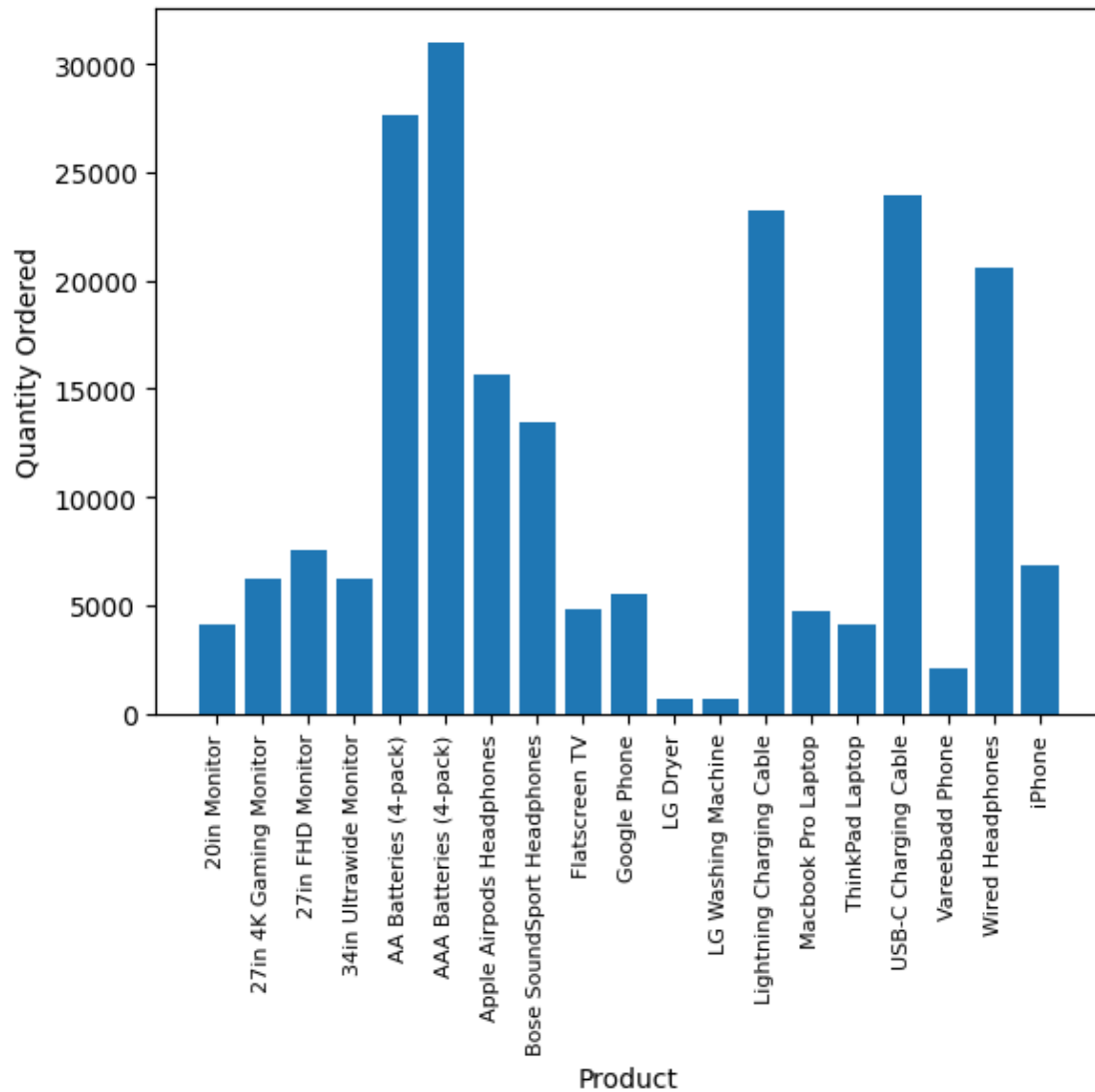

```
[22]: hours = sorted(all_data["Hour"].unique())
plt.plot(hours, all_data.groupby(["Hour"]).count())
plt.xticks(hours)
plt.grid()
plt.xlabel("Hour")
plt.ylabel("Count of Orders")
plt.show()
```



#Highest number of orders came on 11 AM & 7 PM (11)

0.0.7 What products sold the most? Why do you think it sold the most?

```
[23]: product_group = all_data.groupby("Product")
quantity_order = product_group["Quantity Ordered"].sum()
products = [product for product, f in product_group]
plt.bar(products, quantity_order)
plt.ylabel("Quantity Ordered")
plt.xlabel("Product")
plt.xticks(products, rotation="vertical", size=8)
plt.show()
```



```
[44]: price = all_data.groupby("Product")["Price Each"].mean()

fig,ax1 = plt.subplots()

ax2 = ax1.twinx()                                # share x axis and create second y axis,
↪useful when we show different y axis in common x axis

ax1.bar(products,quantity_order, color="green")
ax2.plot(products,price)

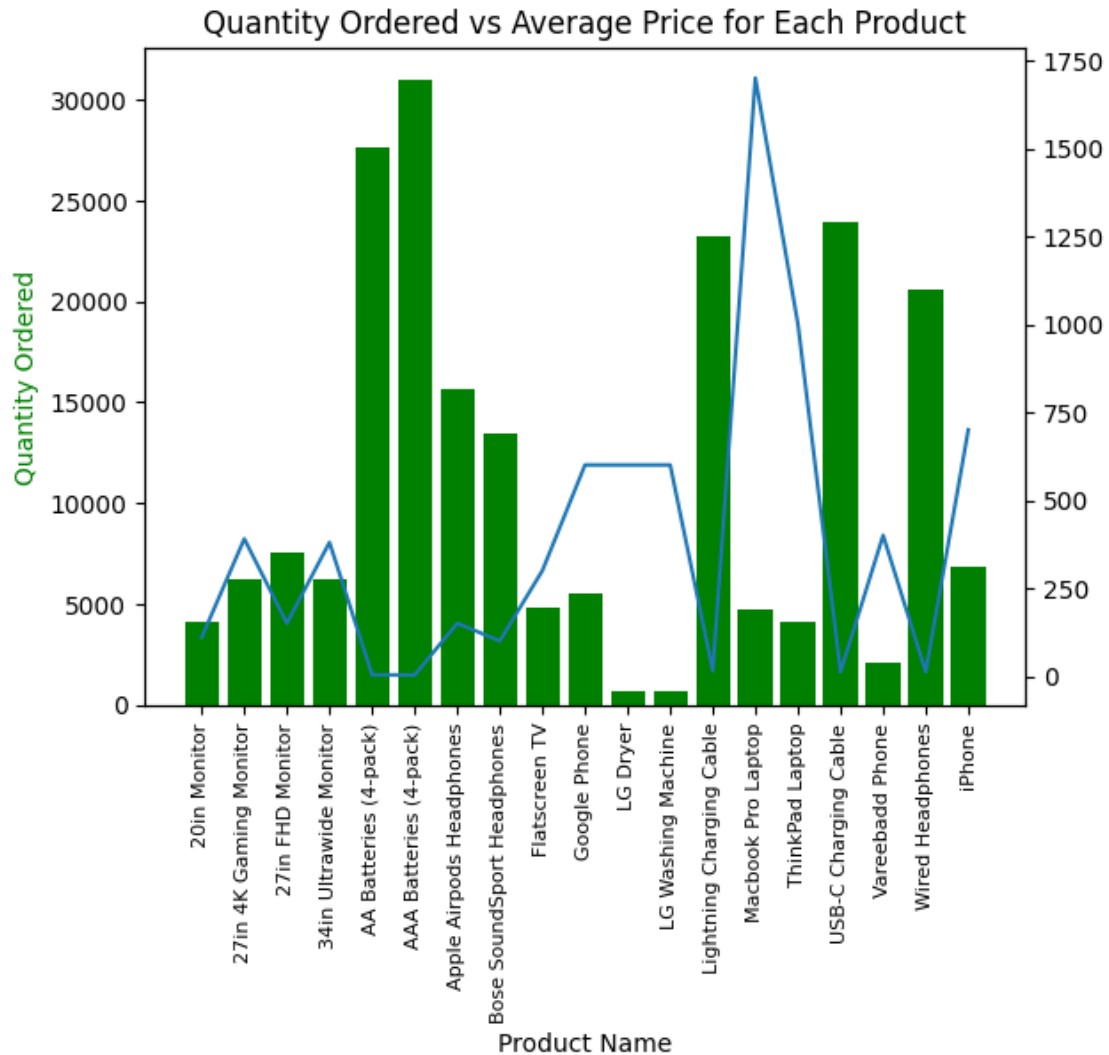
ax1.set_xlabel("Product Name")
ax1.set_ylabel("Quantity Ordered", color="green")
```

```
ax1.set_xticklabels(products, rotation="vertical", size=8)
plt.title("Quantity Ordered vs Average Price for Each Product")

plt.show()
```

C:\Users\DELL\AppData\Local\Temp\ipykernel_9116\836746615.py:13: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.

```
ax1.set_xticklabels(products, rotation="vertical", size=8)
```



#The products with lower prices sold more, while those with higher prices sold less.